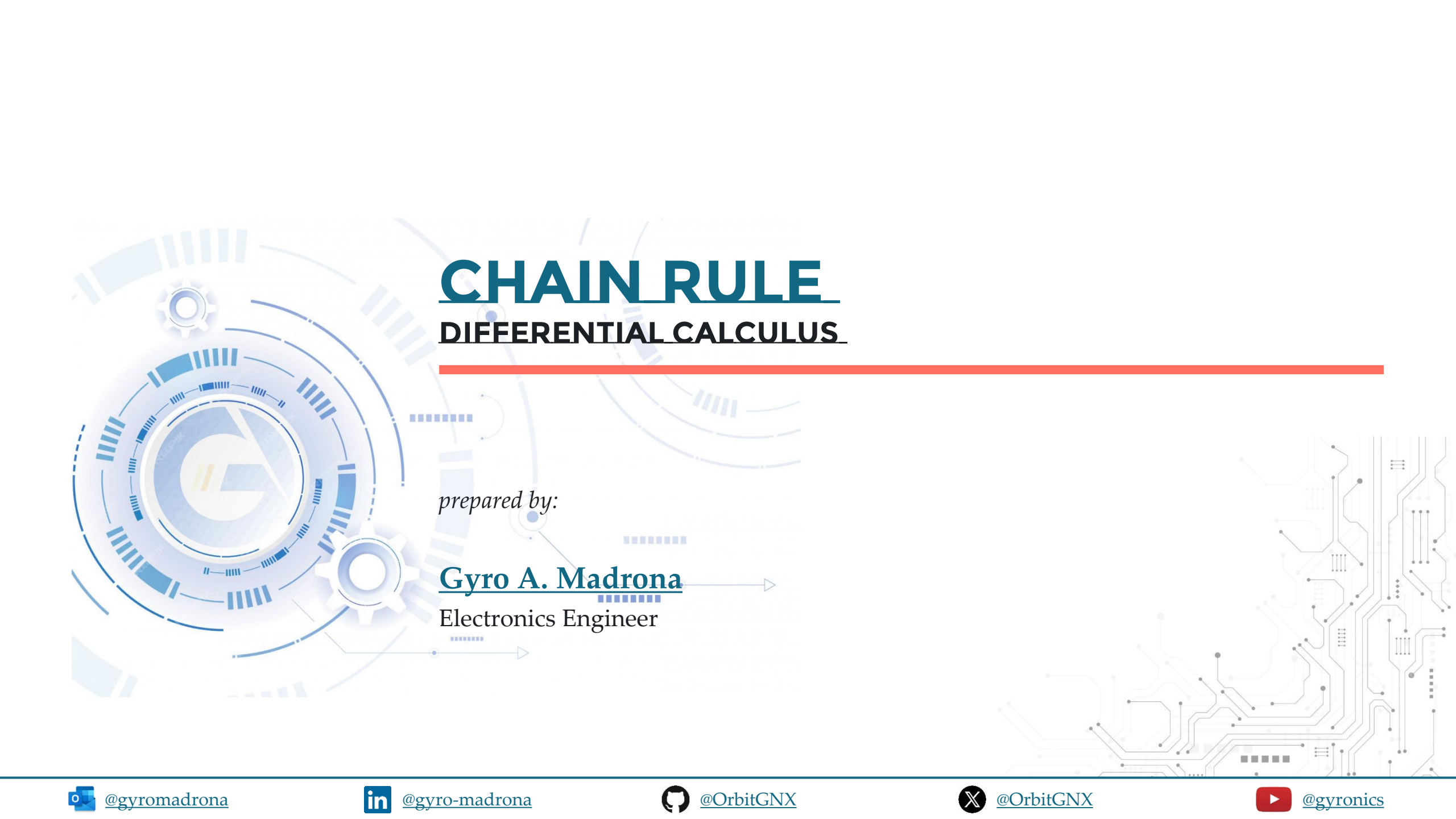




CHAIN RULE

DIFFERENTIAL CALCULUS

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TOPIC OUTLINE

Chain Rule

- Leibniz Notation



CHAIN RULE



CHAIN RULE

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function defined by

$$F(x) = f(g(x))$$

is differentiable at x and F' is given by product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

Differentiate the function $y = \sin(2x)$.
Let $g = 2x \rightarrow g' = 2$

$$f = \sin(g) \rightarrow f' = \cos(g)$$

$$y' = f'(g) \cdot g'$$

$$y' = \cos(g) \cdot 2$$

$$y' = 2 \cos(2x)$$

ans

CHAIN RULE

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function defined by

$$F(x) = f(g(x))$$

is differentiable at x and F' is given by product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

Differentiate the function $y = \cos(\sqrt{x})$.
outer $\swarrow$ $\nwarrow$ inner $g(x)$

$$\text{Let } g = \sqrt{x} \rightarrow g' = \frac{1}{2\sqrt{x}}$$

$$f(g) = \cos(g) \rightarrow f'(g) = -\sin(g)$$

$$y' = f'(g) \cdot g'$$

$$y' = -\sin(g) \cdot g'$$

$$y' = -\frac{1}{2\sqrt{x}} \sin(\sqrt{x})$$

ans



EXERCISE

Differentiate the function $y = \sqrt{x^2 + 1}$.
 $\nwarrow g(x)$

$$\text{Let } g = x^2 + 1 \rightarrow g' = 2x$$

$$f(g) = \sqrt{g} \rightarrow f'(g) = \frac{1}{2\sqrt{g}}$$

Solution

$$y' = f'(g) \cdot g'$$

$$y' = \frac{1}{2\sqrt{g}} \cdot 2x$$

$$y' = \frac{x}{\sqrt{g}}$$

$$y' = \frac{x}{\sqrt{x^2 + 1}}$$

ans



EXERCISE

Differentiate the function $y = \tan(5x^3 + 2x)$.
 $g(x)$

$$\text{let } g = 5x^3 + 2x \rightarrow g' = 15x^2 + 2$$

$$f(g) = \tan(g) \rightarrow f'(g) = \sec^2(g)$$

Solution

$$y' = f'(g) \cdot g'$$

$$y' = \sec^2(g) \cdot (15x^2 + 2)$$

$$y' = (15x^2 + 2) \sec^2(5x^3 + 2x)$$

ans



LEIBNIZ NOTATION

If $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Differentiate the function $y = \sin(2x)$.

$$\frac{dy}{dx} = \frac{d}{du}(\sin u) \cdot \frac{d}{dx}(2x)$$

$$y' = \cos u \cdot 2$$

$$y' = 2 \cos(2x)$$



LEIBNIZ NOTATION

If $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Differentiate the function $y = \cos(\sqrt{x})$.

$$\frac{dy}{dx} = \frac{d}{du}(\cos u) \frac{d}{dx}(\sqrt{x})$$

$$y' = (-\sin u) \left(\frac{1}{2\sqrt{x}} \right)$$

$$y' = -\frac{1}{2\sqrt{x}} \sin(\sqrt{x})$$

ans



EXERCISE

Differentiate $y = \sin x^2$.

Solution

$$\frac{dy}{dx} = \frac{d}{du}(\sin u) \frac{d}{dx}(x^2)$$

$$y' = (\cos u)(2x)$$

$$y' = 2x \cos(x^2)$$

ans



EXERCISE

Differentiate $y = x \sin \frac{1}{x}$.

Solution

$$\frac{dy}{dx} = x \frac{d}{dx} \left(\sin \frac{1}{x} \right) + \sin \frac{1}{x} \frac{d}{dx} (x) \longrightarrow y = \sin \frac{1}{x}$$

$$y' = x \left[-\frac{1}{x^2} \cos \left(\frac{1}{x} \right) \right] + \sin \left(\frac{1}{x} \right) (1)$$

$$y' = -\frac{1}{x} \cos \left(\frac{1}{x} \right) + \sin \left(\frac{1}{x} \right)$$

ans

$$y' = \frac{d}{du} (\sin u) \frac{d}{dx} \left(\frac{1}{x} \right) \xrightarrow{x^{-1}}$$

$$y' = (\cos u) \left(-\frac{1}{x^2} \right)$$

$$y' = -\frac{1}{x^2} \cos \left(\frac{1}{x} \right)$$



EXERCISE

Differentiate $y = \frac{x^3}{\sqrt{1+x^2}}$

Solution

$$\frac{dy}{dx} = \frac{\sqrt{1+x^2} \frac{d}{dx}(x^3) - x^3 \frac{d}{dx}(\sqrt{1+x^2})}{(\sqrt{1+x^2})^2} \rightarrow$$

$$y' = \frac{\sqrt{1+x^2} (3x^2) - x^3 \left(\frac{x}{\sqrt{1+x^2}} \right)}{1+x^2}$$

$$y' = \frac{3x^2 \sqrt{1+x^2} - \frac{x^4}{\sqrt{1+x^2}}}{1+x^2}$$

$$y = \sqrt{1+x^2}$$

$$y' = \frac{d}{du}(\sqrt{u}) \frac{d}{dx}(1+x^2)$$

$$y' = \frac{1}{2\sqrt{u}} (2x)$$

$$y' = \frac{x}{\sqrt{1+x^2}}$$



EXERCISE

Differentiate $y = \frac{x^3}{\sqrt{1+x^2}}$.

$$y' = \frac{3x^2 \cancel{\sqrt{1+x^2}} \cdot \frac{\cancel{\sqrt{1+x^2}}}{\sqrt{1+x^2}} - \frac{x^4}{\sqrt{1+x^2}}}{1+x^2}$$

$$y' = \frac{3x^2(1+x^2) - x^4}{\sqrt{1+x^2}}$$

Solution

$$y' = \frac{3x^2 + 3x^4 - x^4}{(1+x^2)^{1/2}(1+x^2)}$$

$$y' = \frac{2x^4 + 3x^2}{(1+x^2)^{3/2}}$$

Ans



EXERCISE

Differentiate $g(t) = \left(\frac{t-2}{2t+1}\right)^9$.

$$g(t)' = \frac{d}{du} (u^9) \frac{d}{dt} \left(\frac{t-2}{2t+1}\right) \rightarrow$$

$$g(t)' = 9u^8 \cdot \frac{1}{(2t+1)^2}$$

$$g(t)' = 9 \left(\frac{t-2}{2t+1}\right)^8 \cdot \frac{1}{(2t+1)^2}$$

$$g(t)' = \frac{9(t-2)^8}{(2t+1)^{10}}$$

ans

Solution

$$y = \frac{t-2}{2t+1}$$

$$y' = \frac{(2t+1) \frac{d}{dt}(t-2) - (t-2) \frac{d}{dt}(2t+1)}{(2t+1)^2}$$

$$y' = \frac{(2t+1)(1) - (t-2)(2)}{(2t+1)^2}$$

$$y' = \frac{(2t+1) - (2t-4)}{(2t+1)^2} \leftarrow t$$



EXERCISE

The displacement of a particle on a vibrating string is given by the equation

$$s(t) = 10 + \frac{1}{4} \sin(10\pi t)$$

where s is measured in centimeters and t in seconds.

Find the **velocity** of the particle in t seconds.

$$\frac{ds(t)}{dt}$$

Solution

$$s(t) = 10 + \frac{1}{4} \sin(u)$$

$$\frac{ds(t)}{dt} = \frac{d}{dt}(10) + \frac{1}{4} \frac{d}{du} \sin(u) \frac{d}{dt}(10\pi t)$$

$$s'(t) = \frac{1}{4} \cos(u) (10\pi)$$

$$s'(t) = \frac{10\pi}{4} \cos(10\pi t)$$

$$s'(t) = \frac{5}{2} \pi \cos(10\pi t)$$

ans



EXERCISE

Differentiate $y = \frac{\cos \pi x}{\sin \pi x + \cos \pi x}$.

Solution

$$\frac{dy}{dx} = \frac{(\sin \pi x + \cos \pi x) \frac{d}{dx}(\cos \pi x) - \cos \pi x \frac{d}{dx}(\sin \pi x + \cos \pi x)}{(\sin \pi x + \cos \pi x)^2}$$

$$y' = \frac{(\sin \pi x + \cos \pi x)(-\pi \sin \pi x) - \cos \pi x (\pi \cos \pi x - \pi \sin \pi x)}{(\sin \pi x + \cos \pi x)^2}$$

$$y' = \frac{-\pi \sin^2(\pi x) - \pi \sin(\pi x) \cos(\pi x) - \pi \cos^2(\pi x) + \pi \cos(\pi x) \sin(\pi x)}{(\sin \pi x + \cos \pi x)^2}$$



EXERCISE

Differentiate $y = \frac{\cos \pi x}{\sin \pi x + \cos \pi x}$.

Solution

$$y' = \frac{-\pi [\cancel{\sin^2(\pi x)} + \cancel{\cos^2(\pi x)}]}{(\sin \pi x + \cos \pi x)^2}$$

$$y' = -\frac{\pi}{(\sin \pi x + \cos \pi x)^2}$$

ans



EXERCISE

A Cepheid variable star is a star whose brightness alternately increases and decreases. The most easily visible such star is Delta Cephei, for which the interval between times of maximum brightness is 5.4 days. The average brightness of this star is 4.0 and its brightness changes by ± 0.35 . In view of these data, the brightness of Delta Cephei at time t , where t is measured in days, has been modeled by the function

$$B(t) = 4.0 + 0.35 \sin\left(\frac{2\pi t}{5.4}\right)$$

Find the **rate** of change of the **brightness** after t days.

$$\frac{dB(t)}{dt}$$

Solution

$$B(t) = 4.0 + 0.35 \sin\left(\frac{2\pi t}{5.4}\right)$$

$$B'(t) = \frac{d}{dt}(4.0) + 0.35 \left[\frac{d}{du} \sin u \cdot \frac{d}{dt} \left(\frac{2\pi}{5.4} t \right) \right]$$

$$B'(t) = 0.35 \cos u \cdot \frac{2\pi}{5.4}$$

$$B'(t) = \frac{7\pi}{5.4} \cos\left(\frac{2\pi t}{5.4}\right)$$

ans



LABORATORY

